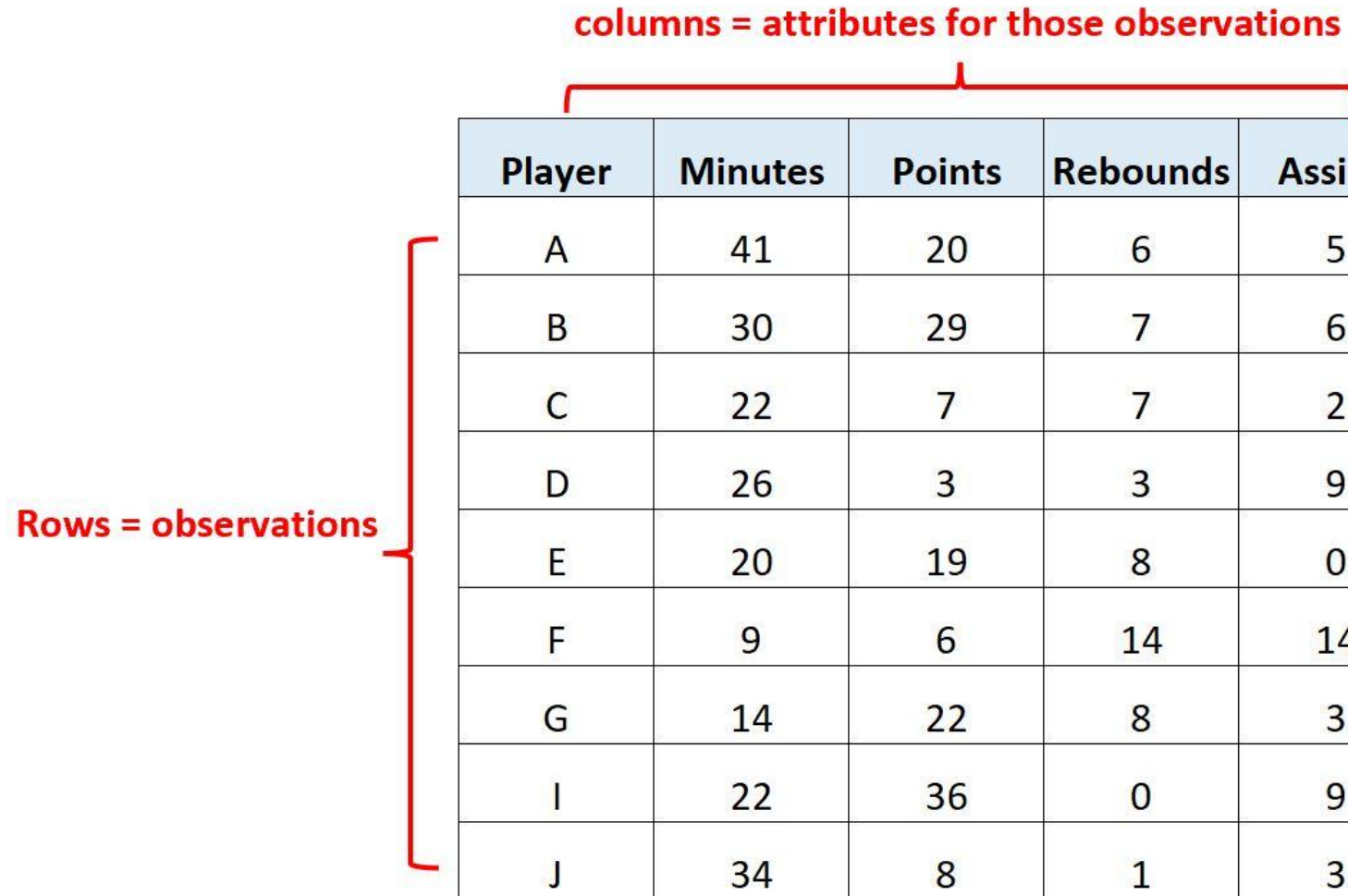


Tabular Data

columns = attributes for those observations

Rows = observations



Player	Minutes	Points	Rebounds	Assists
A	41	20	6	5
B	30	29	7	6
C	22	7	7	2
D	26	3	3	9
E	20	19	8	0
F	9	6	14	14
G	14	22	8	3
I	22	36	0	9
J	34	8	1	3

HANDLING REAL-WORLD TABULAR DATA

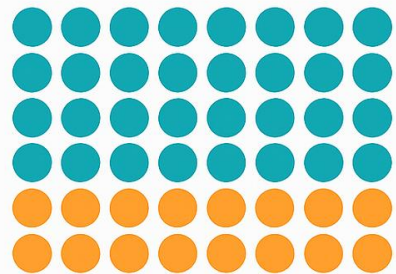
Dealing with Imbalance and Introduction to NLP

Instructor: Zaryab Ahmad Khan

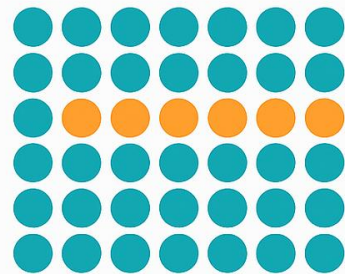
IMBALANCED DATASETS (THE 99% PROBLEM)

IMBALANCED DATASETS HOW TO HANDLE THEM

Imbalanced Dataset



Balanced Dataset



- **The Problem:** In many real-world scenarios, one category is massive and the other is tiny.
- **Example:** If 99% of bank transactions are "Safe" and only 1% are "Fraud," an AI can get 99% accuracy just by guessing "Safe" every time.
- **The Danger:** The model becomes "lazy" and ignores the rare, but important, minority class.

CLASS WEIGHTS (THE BALANCING ACT)

Actual Values	Predicted Prob.(Class 1)	Penalty Cost
0	0.32	0.211
0	0.18	0.109
0	0.28	0.18
0	0.12	0.069
0	0.08	0.045
1	0.48	4.104
0	0.24	0.15
0	0.01	0.005
0	0.22	0.136
0	0.16	0.095

- **The Solution:** We tell the algorithm that the minority class is more important.
- **How it works:** We assign a higher "weight" to the rare class.
- **Effect:** The model gets "punished" more heavily for missing a minority case, forcing it to pay closer attention to those tiny patterns.

Hello, I'm Ironman. I have Friday AI

One Hot Encoding
→

AI	1	0	0	0	0	0	0
Ironman	0	1	0	0	0	0	0
Friday	0	0	1	0	0	0	0
have	0	0	0	1	0	0	0
Hello	0	0	0	0	1	0	0
I	0	0	0	0	0	1	0
I'm	0	0	0	0	0	0	1

BASIC NLP (TURNING WORDS INTO NUMBERS)

- **The Problem:** AI models are math engines; they cannot process a paragraph of text directly.
- **The Solution:** CountVectorizer.
- **Process:** It counts how many times each word appears in a sentence and turns that sentence into a row of numbers.
- **Use Case:** Turning customer reviews or emails into data that a standard model (like Logistic Regression) can read.